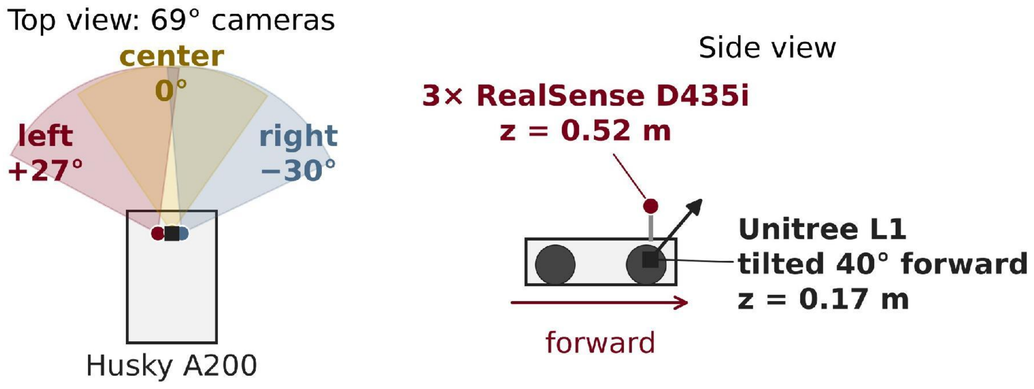
Minnesota Robotics Institute

From Rows to Ground Truth: A Field Robot for Vineyard Mapping and Navigation Learning

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Sensing & localization

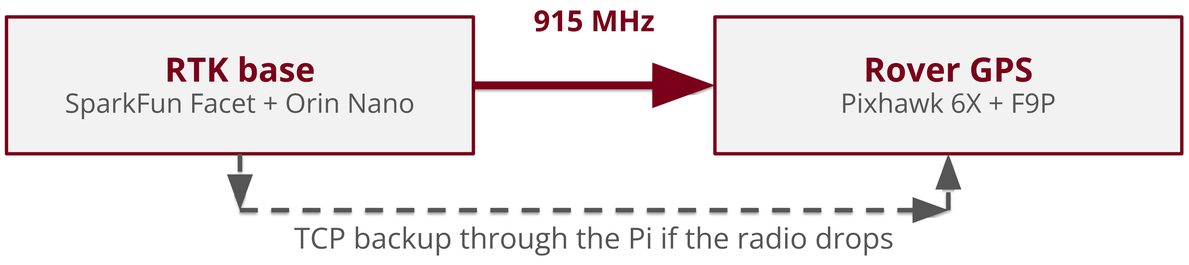


cuVSLAM	FAST-LIO2	Wheel odom
3 camera IR pairs	L1 lidar + IMU	Husky A200

- Visual, lidar and wheel odometry run side by side

Ground-truth data capture

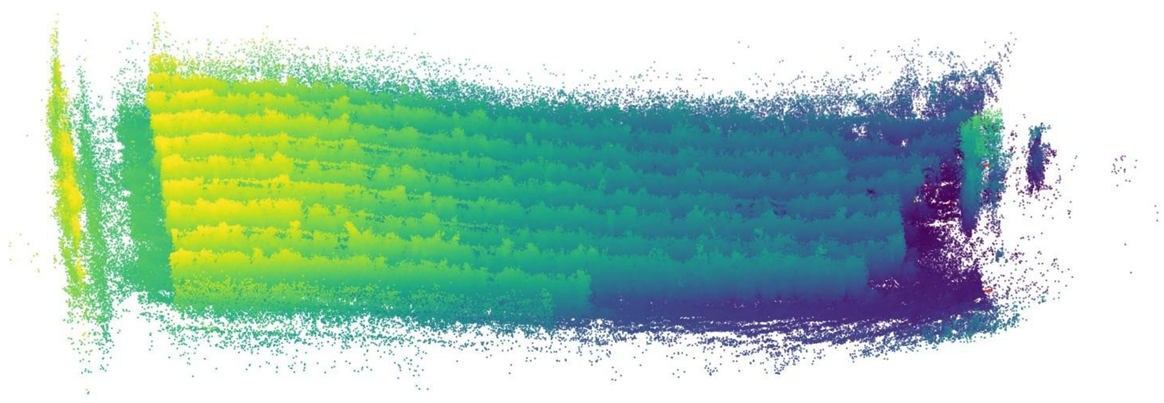
- RTK fixed: 14 mm horizontal, 10 mm vertical



Cameras	colour + 3 IR (JPEG), lossless depth, 3 IMUs
Lidar, GPS	L1 cloud + IMU 10 Hz, RTK state, FCU IMU
Husky	wheel odometry 19 Hz, TF frames
Calibration	camera rotations agree within 0.6° across 3 methods; lidar tilt within 0.25°
Camera timing	3 cameras offset by 4.2 ms and about 8 ms (frame = 16.7 ms)

- Recorder refuses to start if a sensor is missing

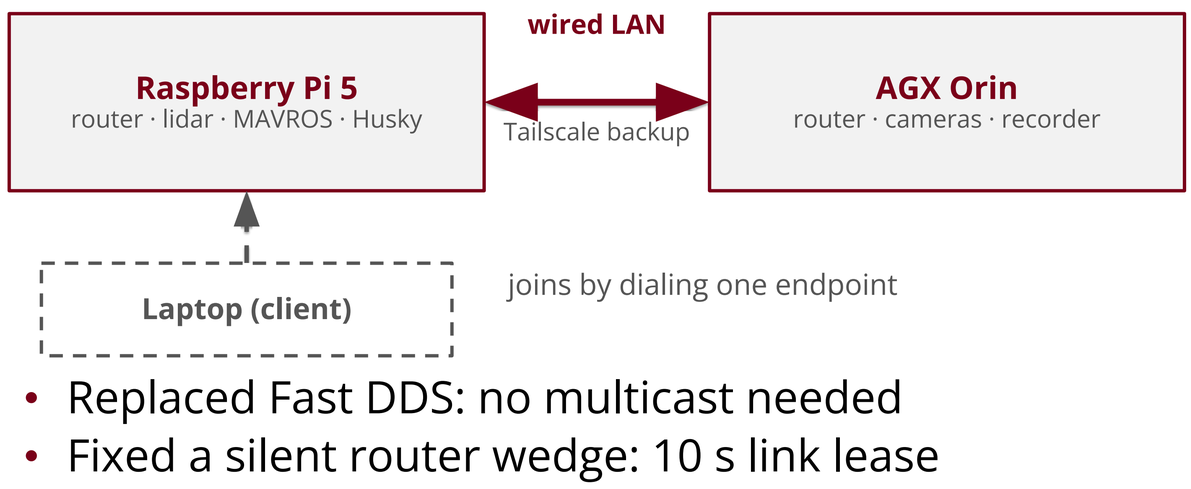
One pass, one vineyard map



4.47 M points · 13,733 keyframes · 537 m · 23 min

What is new

One ROS 2 graph across boards, over Zenoh



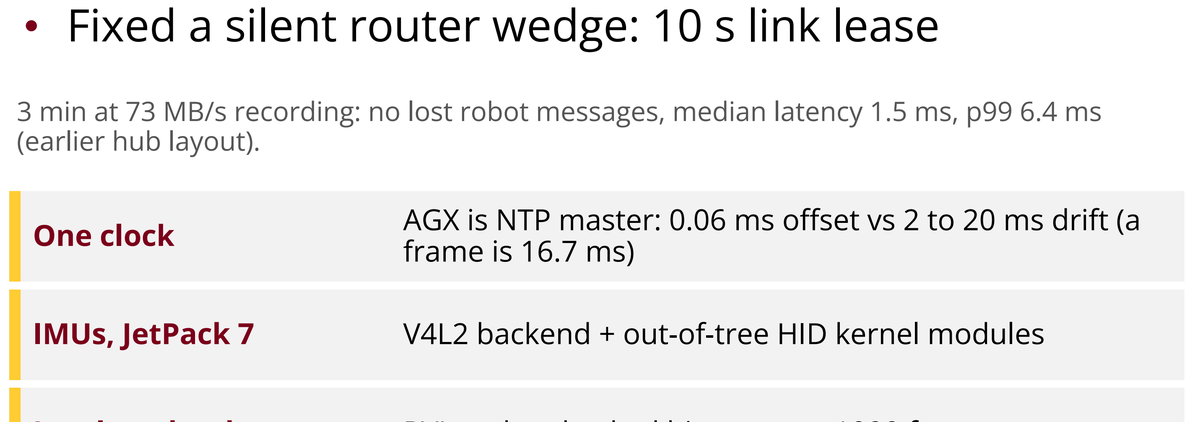
- Replaced Fast DDS: no multicast needed
- Fixed a silent router wedge: 10 s link lease

3 min at 73 MB/s recording: no lost robot messages, median latency 1.5 ms, p99 6.4 ms (earlier hub layout).

One clock	AGX is NTP master: 0.06 ms offset vs 2 to 20 ms drift (a frame is 16.7 ms)
IMUs, JetPack 7	V4L2 backend + out-of-tree HID kernel modules
Lossless depth	RVL codec checked bit-exact on 1000 frames

Navigation policy

- Sees 3 camera frames, a lidar height map, a goal and speed
- DINOv3 + transformer + diffusion head: 8 waypoints over 4 s
- Trained on vineyard drives; a whole vineyard held out
- Half the error of simple baselines on sharp turns



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